

BIMBINGAN TUGAS AKHIR #5

Jumat, 9 Oktober 2026

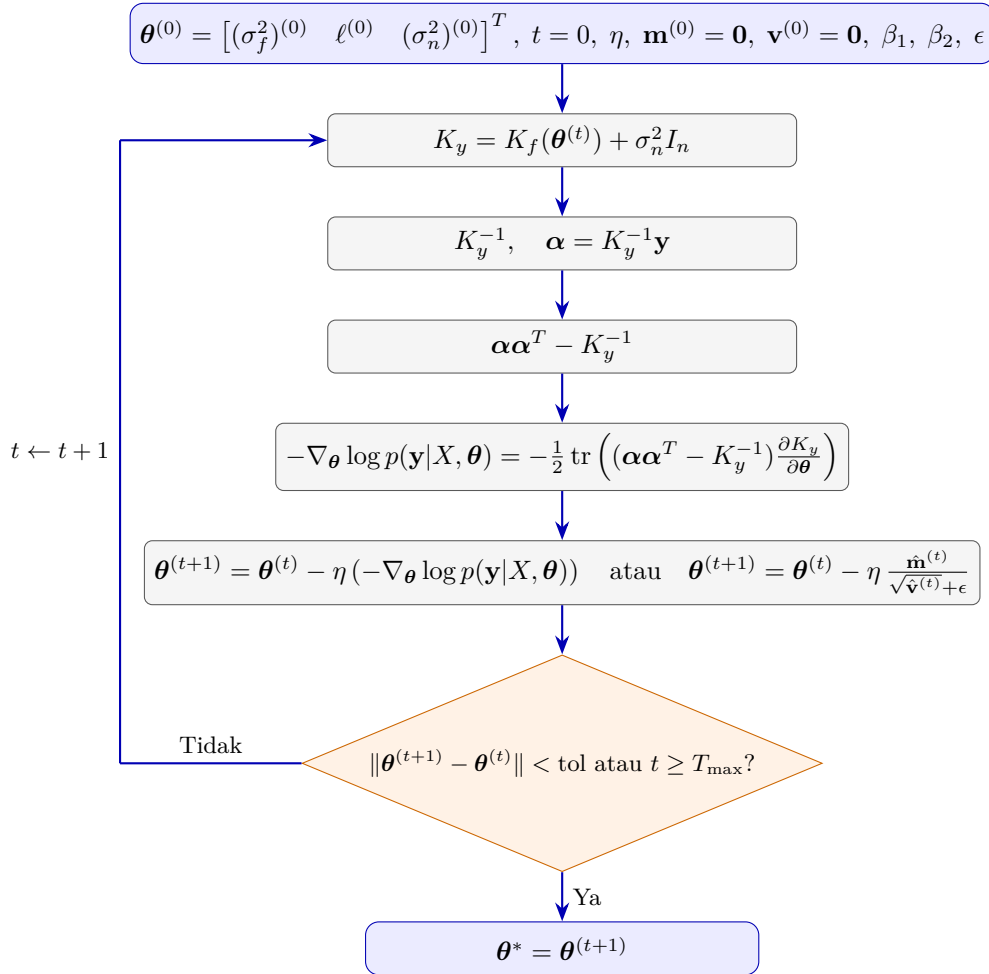
1 Contoh Perhitungan Optimasi Log Marginal Likelihood (2×2)

(Rasmussen & Williams, 2006, Bab 5.4.1, hlm. 113–114, Persamaan (5.9))

$$\frac{\partial \log p(\mathbf{y}|X, \boldsymbol{\theta})}{\partial \theta_j} = \frac{1}{2} \text{tr} \left((\boldsymbol{\alpha} \boldsymbol{\alpha}^T - K_y^{-1}) \frac{\partial K_y}{\partial \theta_j} \right), \quad \text{dengan } \boldsymbol{\alpha} = K_y^{-1} \mathbf{y} \quad (1)$$

Karena algoritma optimasi (seperti *Gradient Descent* maupun *Adam*) bertujuan untuk meminimumkan suatu fungsi objektif sedangkan tujuan kita adalah memaksimumkan *log marginal likelihood*, maka fungsi objektif yang diminimumkan dikalikan -1 ($-\log p(\mathbf{y}|X, \boldsymbol{\theta})$):

$$-\frac{\partial \log p(\mathbf{y}|X, \boldsymbol{\theta})}{\partial \theta_j} = -\frac{1}{2} \text{tr} \left((\boldsymbol{\alpha} \boldsymbol{\alpha}^T - K_y^{-1}) \frac{\partial K_y}{\partial \theta_j} \right) \quad (2)$$



Gambar 1: Diagram Iterasi Optimasi Hyperparameter.

1.1 Gradient Descent

1. Data Observasi & Parameter Awal

- Data Input: $X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, Target: $\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$
- Jarak: $r_{12} = |x_1 - x_2| = 1 \implies r_{12}^2 = (x_1 - x_2)^2 = 1$
- Kernel RBF: $k(x_p, x_q) = \sigma_f^2 \exp\left(-\frac{r_{pq}^2}{2\ell^2}\right)$
- Inisialisasi: $\boldsymbol{\theta}^{(0)} = \begin{bmatrix} (\sigma_f^2)^{(0)} \\ \ell^{(0)} \\ (\sigma_n^2)^{(0)} \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0,1 \end{bmatrix}$, Learning Rate: $\eta = 0,1$

2. Matriks Kovariansi K_y

$$k_{11} = k_{22} = \sigma_f^2 = 1 \quad (3)$$

$$k_{12} = k_{21} = 1 \cdot \exp\left(-\frac{1}{2(1)^2}\right) = e^{-0,5} \approx 0,6065 \quad (4)$$

$$K_y = \begin{bmatrix} \sigma_f^2 + \sigma_n^2 & k_{12} \\ k_{21} & \sigma_f^2 + \sigma_n^2 \end{bmatrix} = \begin{bmatrix} 1 + 0,1 & 0,6065 \\ 0,6065 & 1 + 0,1 \end{bmatrix} = \begin{bmatrix} 1,1 & 0,6065 \\ 0,6065 & 1,1 \end{bmatrix} \quad (5)$$

3. Determinan, Invers K_y^{-1} , dan Vektor α

$$|K_y| = (1,1)^2 - (0,6065)^2 = 1,21 - 0,3678 = 0,8422 \quad (6)$$

$$K_y^{-1} = \frac{1}{0,8422} \begin{bmatrix} 1,1 & -0,6065 \\ -0,6065 & 1,1 \end{bmatrix} = \begin{bmatrix} 1,3062 & -0,7202 \\ -0,7202 & 1,3062 \end{bmatrix} \quad (7)$$

$$\alpha = K_y^{-1} \mathbf{y} = \begin{bmatrix} 1,3062 & -0,7202 \\ -0,7202 & 1,3062 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} (1,3062)(1) + (-0,7202)(2) \\ (-0,7202)(1) + (1,3062)(2) \end{bmatrix} = \begin{bmatrix} -0,1342 \\ 1,8922 \end{bmatrix} \quad (8)$$

4. Matriks $(\alpha\alpha^T - K_y^{-1})$

$$\begin{aligned} \alpha\alpha^T &= \begin{bmatrix} -0,1342 \\ 1,8922 \end{bmatrix} \begin{bmatrix} -0,1342 & 1,8922 \end{bmatrix} \\ &= \begin{bmatrix} (-0,1342)^2 & (-0,1342)(1,8922) \\ (1,8922)(-0,1342) & (1,8922)^2 \end{bmatrix} = \begin{bmatrix} 0,0180 & -0,2540 \\ -0,2540 & 3,5805 \end{bmatrix} \end{aligned} \quad (9)$$

$$\alpha\alpha^T - K_y^{-1} = \begin{bmatrix} 0,0180 & -0,2540 \\ -0,2540 & 3,5805 \end{bmatrix} - \begin{bmatrix} 1,3062 & -0,7202 \\ -0,7202 & 1,3062 \end{bmatrix} = \begin{bmatrix} -1,2882 & 0,4662 \\ 0,4662 & 2,2743 \end{bmatrix} \quad (10)$$

5. Perhitungan Gradien

Bentuk umum matriks kovariansi K_y sebagai fungsi dari hyperparameter $(\sigma_f^2, \ell, \sigma_n^2)$ dengan $r_{12}^2 = 1$:

$$K_y = \begin{bmatrix} \sigma_f^2 + \sigma_n^2 & \sigma_f^2 \exp\left(-\frac{1}{2\ell^2}\right) \\ \sigma_f^2 \exp\left(-\frac{1}{2\ell^2}\right) & \sigma_f^2 + \sigma_n^2 \end{bmatrix} \quad (11)$$

a. Terhadap σ_f^2 :

$$\frac{\partial K_y}{\partial \sigma_f^2} = \begin{bmatrix} 1 & 0,6065 \\ 0,6065 & 1 \end{bmatrix} \quad (12)$$

$$(\boldsymbol{\alpha}\boldsymbol{\alpha}^T - K_y^{-1}) \frac{\partial K_y}{\partial \sigma_f^2} = \begin{bmatrix} -1,2882 & 0,4662 \\ 0,4662 & 2,2743 \end{bmatrix} \begin{bmatrix} 1 & 0,6065 \\ 0,6065 & 1 \end{bmatrix} = \begin{bmatrix} -1,0054 & -0,3151 \\ 1,8456 & 2,5570 \end{bmatrix} \quad (13)$$

$$\text{tr} \left((\boldsymbol{\alpha}\boldsymbol{\alpha}^T - K_y^{-1}) \frac{\partial K_y}{\partial \sigma_f^2} \right) = -1,0054 + 2,5570 = 1,5516 \quad (14)$$

$$\frac{\partial \log p(\mathbf{y}|X, \boldsymbol{\theta})}{\partial \sigma_f^2} = \frac{1}{2}(1,5516) = 0,7758 \implies -\frac{\partial \log p(\mathbf{y}|X, \boldsymbol{\theta})}{\partial \sigma_f^2} = -0,7758 \quad (15)$$

b. Terhadap ℓ :

$$\frac{\partial K_y}{\partial \ell} = \begin{bmatrix} 0 & 0,6065 \cdot \frac{r_{12}^2}{1^3} \\ 0,6065 \cdot \frac{r_{12}^2}{1^3} & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0,6065 \\ 0,6065 & 0 \end{bmatrix} \quad (16)$$

$$(\boldsymbol{\alpha}\boldsymbol{\alpha}^T - K_y^{-1}) \frac{\partial K_y}{\partial \ell} = \begin{bmatrix} -1,2882 & 0,4662 \\ 0,4662 & 2,2743 \end{bmatrix} \begin{bmatrix} 0 & 0,6065 \\ 0,6065 & 0 \end{bmatrix} = \begin{bmatrix} 0,2828 & -0,7813 \\ 1,3794 & 0,2828 \end{bmatrix} \quad (17)$$

$$\text{tr} \left((\boldsymbol{\alpha}\boldsymbol{\alpha}^T - K_y^{-1}) \frac{\partial K_y}{\partial \ell} \right) = 0,2828 + 0,2828 = 0,5656 \quad (18)$$

$$\frac{\partial \log p(\mathbf{y}|X, \boldsymbol{\theta})}{\partial \ell} = \frac{1}{2}(0,5656) = 0,2828 \implies -\frac{\partial \log p(\mathbf{y}|X, \boldsymbol{\theta})}{\partial \ell} = -0,2828 \quad (19)$$

c. Terhadap σ_n^2 :

$$\frac{\partial K_y}{\partial \sigma_n^2} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2 \quad (20)$$

$$(\boldsymbol{\alpha}\boldsymbol{\alpha}^T - K_y^{-1}) \frac{\partial K_y}{\partial \sigma_n^2} = \begin{bmatrix} -1,2882 & 0,4662 \\ 0,4662 & 2,2743 \end{bmatrix} \quad (21)$$

$$\text{tr} \left((\boldsymbol{\alpha}\boldsymbol{\alpha}^T - K_y^{-1}) \frac{\partial K_y}{\partial \sigma_n^2} \right) = -1,2882 + 2,2743 = 0,9861 \quad (22)$$

$$\frac{\partial \log p(\mathbf{y}|X, \boldsymbol{\theta})}{\partial \sigma_n^2} = \frac{1}{2}(0,9861) = 0,4930 \implies -\frac{\partial \log p(\mathbf{y}|X, \boldsymbol{\theta})}{\partial \sigma_n^2} = -0,4930 \quad (23)$$

6. Update Parameter

$$\nabla_{\boldsymbol{\theta}} (-\log p(\mathbf{y}|X, \boldsymbol{\theta})) = \begin{bmatrix} -\frac{\partial \log p(\mathbf{y}|X, \boldsymbol{\theta})}{\partial \sigma_f^2} \\ -\frac{\partial \log p(\mathbf{y}|X, \boldsymbol{\theta})}{\partial \ell} \\ -\frac{\partial \log p(\mathbf{y}|X, \boldsymbol{\theta})}{\partial \sigma_n^2} \end{bmatrix} = \begin{bmatrix} -0,7758 \\ -0,2828 \\ -0,4930 \end{bmatrix} \quad (24)$$

$$\boldsymbol{\theta}^{(1)} = \boldsymbol{\theta}^{(0)} - \eta \nabla_{\boldsymbol{\theta}} (-\log p(\mathbf{y}|X, \boldsymbol{\theta})) \quad (25)$$

$$\begin{bmatrix} (\sigma_f^2)^{(1)} \\ \ell^{(1)} \\ (\sigma_n^2)^{(1)} \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0,1 \end{bmatrix} - 0,1 \begin{bmatrix} -0,7758 \\ -0,2828 \\ -0,4930 \end{bmatrix} = \begin{bmatrix} 1,0776 \\ 1,0283 \\ 0,1493 \end{bmatrix} \quad (26)$$

1.2 Adam

Inisialisasi: $\eta = 0,1$, $\beta_1 = 0,9$, $\beta_2 = 0,999$, $\epsilon = 10^{-8}$, serta $\mathbf{m}^{(0)} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$, $\mathbf{v}^{(0)} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$.

Menggunakan gradien dari perhitungan sebelumnya:

$$-\nabla_{\boldsymbol{\theta}} (\log p(\mathbf{y}|X, \boldsymbol{\theta})) = \begin{bmatrix} -0,7758 \\ -0,2828 \\ -0,4930 \end{bmatrix}$$

1. Estimasi Momen Pertama & Kedua

$$\begin{aligned} \mathbf{m}^{(1)} &= \beta_1 \mathbf{m}^{(0)} + (1 - \beta_1) (-\nabla_{\boldsymbol{\theta}} \log p(\mathbf{y}|X, \boldsymbol{\theta})) \\ &= 0,9 \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + 0,1 \begin{bmatrix} -0,7758 \\ -0,2828 \\ -0,4930 \end{bmatrix} = \begin{bmatrix} -0,0776 \\ -0,0283 \\ -0,0493 \end{bmatrix} \end{aligned} \quad (27)$$

$$\begin{aligned} \mathbf{v}^{(1)} &= \beta_2 \mathbf{v}^{(0)} + (1 - \beta_2) (-\nabla_{\boldsymbol{\theta}} \log p(\mathbf{y}|X, \boldsymbol{\theta}))^2 \\ &= 0,999 \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + 0,001 \begin{bmatrix} (-0,7758)^2 \\ (-0,2828)^2 \\ (-0,4930)^2 \end{bmatrix} = \begin{bmatrix} 0,000602 \\ 0,00008 \\ 0,000243 \end{bmatrix} \end{aligned} \quad (28)$$

2. Koreksi Bias

$$\hat{\mathbf{m}}^{(1)} = \frac{\mathbf{m}^{(1)}}{1 - \beta_1^1} = \frac{1}{0,1} \begin{bmatrix} -0,0776 \\ -0,0283 \\ -0,0493 \end{bmatrix} = \begin{bmatrix} -0,7758 \\ -0,2828 \\ -0,4930 \end{bmatrix} \quad (29)$$

$$\hat{\mathbf{v}}^{(1)} = \frac{\mathbf{v}^{(1)}}{1 - \beta_2^1} = \frac{1}{0,001} \begin{bmatrix} 0,000602 \\ 0,00008 \\ 0,000243 \end{bmatrix} = \begin{bmatrix} 0,6019 \\ 0,08 \\ 0,243 \end{bmatrix} \quad (30)$$

3. Update Parameter

$$\begin{aligned} \boldsymbol{\theta}^{(1)} &= \boldsymbol{\theta}^{(0)} - \eta \frac{\hat{\mathbf{m}}^{(1)}}{\sqrt{\hat{\mathbf{v}}^{(1)} + \epsilon}} \\ \begin{bmatrix} (\sigma_f^2)^{(1)} \\ \ell^{(1)} \\ (\sigma_n^2)^{(1)} \end{bmatrix} &= \begin{bmatrix} 1 \\ 1 \\ 0,1 \end{bmatrix} - 0,1 \begin{bmatrix} \frac{-0,7758}{\sqrt{0,6019+10^{-8}}} \\ \frac{-0,2828}{\sqrt{0,08+10^{-8}}} \\ \frac{-0,4930}{\sqrt{0,243+10^{-8}}} \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0,1 \end{bmatrix} - 0,1 \begin{bmatrix} -1 \\ -1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1,1 \\ 1,1 \\ 0,2 \end{bmatrix} \end{aligned} \quad (31)$$

2 Eksplorasi Operasi Kombinasi Kernel

Empat kombinasi kernel yang diuji pada eksperimen ini didefinisikan sebagai:

$$k_{\text{RBF}+\text{Per}}(x, x') = \sigma_f^2 \exp\left(-\frac{(x-x')^2}{2\ell_{\text{RBF}}^2}\right) + \sigma_p^2 \exp\left(-\frac{2 \sin^2\left(\frac{\pi|x-x'|}{p}\right)}{\ell_{\text{Per}}^2}\right) \quad (32)$$

$$k_{\text{RBF} \times \text{Per}}(x, x') = \sigma_f^2 \sigma_p^2 \exp\left(-\frac{(x-x')^2}{2\ell_{\text{RBF}}^2} - \frac{2 \sin^2\left(\frac{\pi|x-x'|}{p}\right)}{\ell_{\text{Per}}^2}\right) \quad (33)$$

$$k_{\text{Lin}+\text{Per}}(x, x') = \sigma_b^2 + \sigma_v^2(x-c)(x'-c) + \sigma_p^2 \exp\left(-\frac{2 \sin^2\left(\frac{\pi|x-x'|}{p}\right)}{\ell_{\text{Per}}^2}\right) \quad (34)$$

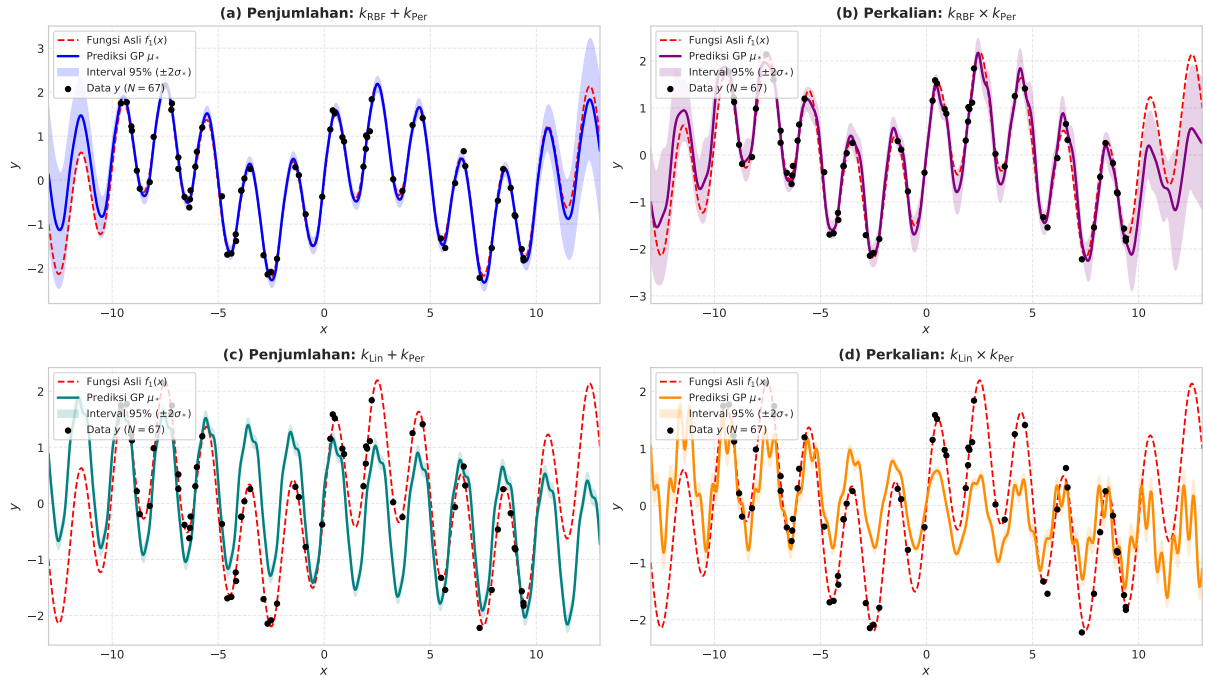
$$k_{\text{Lin} \times \text{Per}}(x, x') = \left[\sigma_b^2 + \sigma_v^2(x-c)(x'-c)\right] \times \sigma_p^2 \exp\left(-\frac{2 \sin^2\left(\frac{\pi|x-x'|}{p}\right)}{\ell_{\text{Per}}^2}\right) \quad (35)$$

2.1 Eksperimen 1

Fungsi asli yang digunakan:

$$f_1(x) = \sin(0,6x) + 1,2 \sin(\pi x) \quad (36)$$

dengan $N = 67$ titik data observasi $y_i = f_1(x_i) + \varepsilon_i$, $\varepsilon_i \sim \mathcal{N}(0, 0,15^2)$ pada rentang $x \in [-10, 10]$.



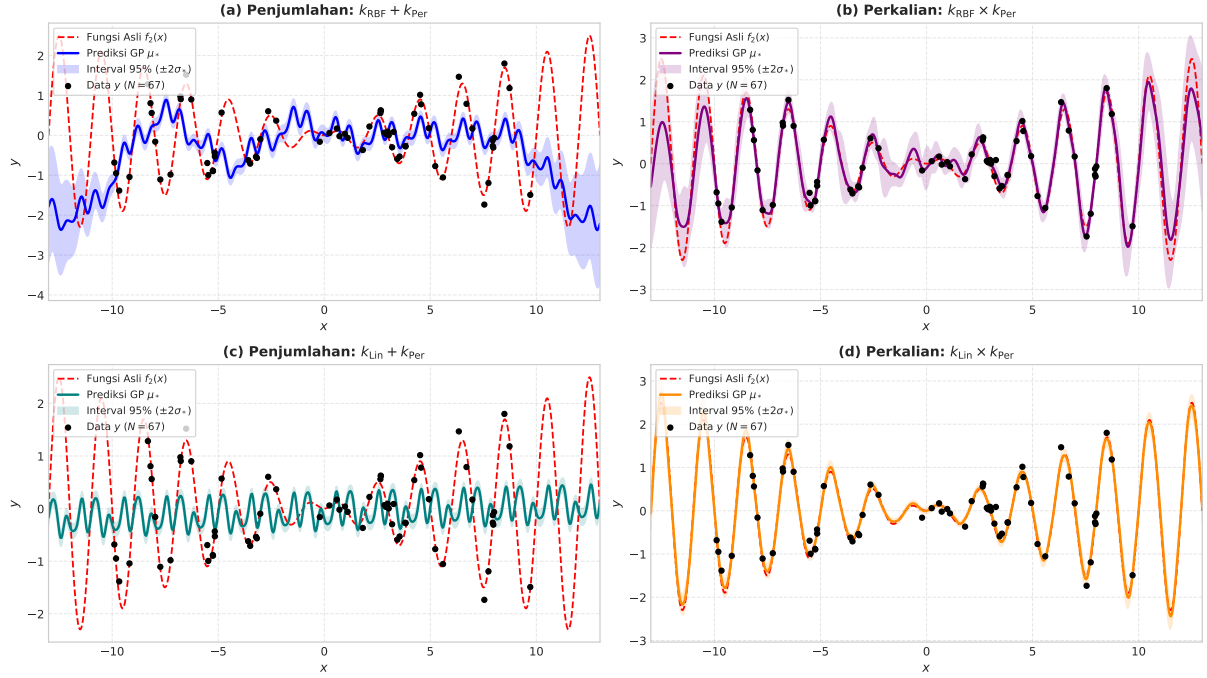
Gambar 2: Hasil *fitting* Gaussian Process Regression pada fungsi asli $f_1(x)$ (36): (a) $k_{\text{RBF}} + k_{\text{Per}}$; (b) $k_{\text{RBF}} \times k_{\text{Per}}$; (c) $k_{\text{Lin}} + k_{\text{Per}}$; (d) $k_{\text{Lin}} \times k_{\text{Per}}$.

2.2 Eksperimen 2

Fungsi asli yang digunakan:

$$f_2(x) = 0,2x \sin(\pi x) \quad (37)$$

dengan $N = 67$ titik data observasi $y_i = f_2(x_i) + \varepsilon_i$, $\varepsilon_i \sim \mathcal{N}(0, 0,15^2)$ pada rentang $x \in [-10, 10]$.



Gambar 3: Hasil *fitting* Gaussian Process Regression pada fungsi asli $f_2(x)$ (37): (a) $k_{\text{RBF}} + k_{\text{Per}}$; (b) $k_{\text{RBF}} \times k_{\text{Per}}$; (c) $k_{\text{Lin}} + k_{\text{Per}}$; (d) $k_{\text{Lin}} \times k_{\text{Per}}$.

3 Eksplorasi Package Python

3.1 `scipy.optimize.minimize`

Fungsi `scipy.optimize.minimize` digunakan untuk mencari nilai parameter atau vektor variabel $\mathbf{x}^*$ yang meminimumkan suatu fungsi objektif skalar $f(\mathbf{x})$:

```
scipy.optimize.minimize(fun, x0, args=(), method=None, jac=None, bounds=None,
                        tol=None, options=None)
```

Parameter-parameter utamanya:

- **fun**: Fungsi objektif skalar $f(\mathbf{x})$ yang akan diminimumkan.
- **x0**: Nilai tebakan awal variabel parameter dalam bentuk array 1D ($\mathbf{x}_0$).
- **args**: Tuple argumen tambahan yang diteruskan ke fungsi objektif **fun** (dan **jac**).
- **method**: Algoritma optimasi yang digunakan, antara lain: 'Nelder-Mead', 'Powell', 'CG', 'BFGS', 'Newton-CG', 'L-BFGS-B', 'TNC', 'COBYLA', 'SLSQP', 'trust-constr', 'dogleg', 'trust-ncg', 'trust-exact', 'trust-krylov'.
- **jac**: Metode/fungsi untuk menghitung gradien atau vektor Jacobian $\nabla f(\mathbf{x})$. Jika bernilai `None` atau boolean `False`, gradien diestimasi secara numerik (*finite difference*).
- **bounds**: Batasan rentang nilai untuk setiap variabel parameter ($x_{\min}, x_{\max}$).
- **tol**: Batas toleransi konvergensi iterasi.
- **options**: *Dictionary* konfigurasi tambahan (seperti batas iterasi maksimum '`maxiter`' atau penayangan proses '`disp`').

3.2 GPyTorch

3.2.1 Fungsi Kovariansi / Kernel (`gpytorch.kernels`)

- `RBFKernel(ard_num_dims=None, batch_shape=torch.Size([]), active_dims=None, lengthscale_prior=None, lengthscale_constraint=None, eps=1e-06)`
- `MaternKernel(nu=2.5, ard_num_dims=None, batch_shape=torch.Size([]), active_dims=None, lengthscale_prior=None, lengthscale_constraint=None, eps=1e-06)`
- `LinearKernel(num_dimensions=None, offset_prior=None, offset_constraint=None, variance_prior=None, variance_constraint=None, batch_shape=torch.Size([]), active_dims=None)`
- `PeriodicKernel(ard_num_dims=None, batch_shape=torch.Size([]), active_dims=None, period_length_prior=None, period_length_constraint=None, lengthscale_prior=None, lengthscale_constraint=None, eps=1e-06)`

3.2.2 Fungsi Objektif Marginal Likelihood (`gpytorch.mlls`)

- `ExactMarginalLogLikelihood(likelihood, model)`

Pustaka

- [1] Rasmussen, C. E., & Williams, C. K. (2006). *Gaussian Processes for Machine Learning*. MIT Press, Cambridge, MA.
- [2] SciPy Developers. *minimize* — *SciPy v1.18.0 Manual*. Diakses dari: <https://docs.scipy.org/doc/scipy/reference/generated/scipy.optimize.minimize.html>
- [3] GPyTorch Developers. *GPyTorch's documentation* — *GPyTorch 1.15.2 documentation*. Diakses dari: <https://docs.gpytorch.ai/en/stable/>